

Program : Diploma in Cyber Forensics and Information Security	
Course Code : 5283C	Course Title: Internet and Web Application Security
Semester: 5	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

After completion of this course the student will be able to:

- Reveal the underlying in web application.
- Understand the security principles in developing a reliable web application.
- Understand security concepts, security professional roles, and security resources in the context of systems and security development life cycle.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic knowledge in Security threats		Cyber forensics and Security threats	IV
Basic Knowledge in Web Technology		Web Technology Lab	III

Course Outcomes :

CO _n	Description	Duration (Hours)	Cognitive level
CO1	Understand Internet security and applications.	16	Understanding
CO2	Understand web application security fundamentals and browser security principles.	14	Understanding
CO3	Understand Web Application Vulnerabilities.	12	Understanding
CO4	Exposure to various security threats to web applications/servers and providing security to web servers.	16	Understanding
	SeriesTest	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Understand Internet security and applications.		
M1.01	Introduction to security, Types of security, Authentication & Authorization	4	Understanding
M1.02	Firewalls, Encryption & Decryption, SSL	4	Understanding
M1.03	Internet Application - E-Mail, Email protocols, Telnet	2	Understanding
M1.04	FTP, Newsgroup, Chat room	3	Understanding
M1.05	Internet Relay Chat, Video Conferencing	3	Understanding
Contents :Introduction to internet security, Types of security, Authentication and authorization, Firewalls, Encryption & Decryption, SSL, Internet Application - E-Mail, Email protocols, Telnet, FTP, Newsgroup, Chat room, Internet Relay Chat, Video Conferencing			
CO2	Understand web application security fundamentals and browser security principles.		
M2.01	Input Validation	2	Understanding
M2.02	Attack Surface Reduction Rules of Thumb	2	Understanding
M2.03	Classifying and Prioritizing Threads	2	Understanding
M2.04	Origin Policy - Exceptions to the Same-Origin Policy	2	Understanding

M2.05	Cross-Site Scripting and Cross-Site Request Forgery	3	Understanding
M2.06	Reflected XSS - HTML Injection	3	Understanding
	SeriesTest–I	1	
Contents: Input Validation – Attack Surface Reduction Rules of Thumb - Classifying and Prioritizing Threads, Origin Policy - Exceptions to the Same-Origin Policy - Cross-Site Scripting and Cross-Site Request Forgery - Reflected XSS - HTML Injection			
CO3	To understand Web Application Vulnerabilities.		
M3.01	Client state manipulation, cookie based attacks, SQL injection, cross domain attack (XSS/XSRF/XSSI), http header injection	4	Understanding
M3.02	SSL vulnerabilities and testing	3	Understanding
M3.03	Proper encryption use in web application.	2	Understanding
M3.04	Session vulnerabilities and testing – Common website weakness	3	Applying
Contents : Client state manipulation, cookie based attacks, SQL injection, , cross domain attack (XSS/XSRF/XSSI) http header injection, SSL vulnerabilities and testing –Proper encryption use in web application – Session vulnerabilities and testing - Common website weakness.			
CO4	Exposure to various security threats to web applications/ servers and providing security to web servers.		
M4.01	Explain web application security threats and defense mechanisms	6	Understanding
M4.02	Illustrate the mechanism of transmitting data	4	Understanding
M4.03	Explain vulnerabilities of access control and securing mechanisms.	6	Understanding
	SeriesTest–II	1	
Contents : Web application security- Key Problem factors – Core defense mechanisms- Handling user access- handling user input- Handling attackers – web spidering – Discovering hidden content. Transmitting data via the client – Hidden form fields – HTTP cookies – URL parameters – Handling client-side data securely – Attacking authentication – design flaws in authentication mechanisms –securing authentication Attacking access controls – Common vulnerabilities – Securing access controls.			

Text/Reference

T/R	Book Title/Author
T2	Sullivan, Bryan and Vincent Liu Web Application Security A Beginner's Guide McGraw Hill Professional 2011.
T3	Stuttard, Dafydd and Marcus Pinto The Web Application Hacker's Handbook: Finding and Exploiting Security Flaws John Wiley Sons 2011
R ₁	Mike Harwood, Ron Price Internet and Web Application Security, 3rd Edition